

## 1. Executive Summary

|                         |         | Traffic Summary        |  | Count      |  |
|-------------------------|---------|------------------------|--|------------|--|
| Metric                  | Value   | Total Flows Observed   |  | 17532      |  |
| Total Threats Mitigated | 15      | Total Packets Analyzed |  | 1848552637 |  |
| High Priority           | 7       | Malicious Packets      |  | 1847621340 |  |
| Low Priority            | 8       | True Negatives Passed  |  | 12700      |  |
| Manual Releases         | 0       |                        |  |            |  |
| Manual Blocks           | 0       | Attack Vector          |  | Count      |  |
|                         |         | ICMP Flood             |  | 12         |  |
| ML Metrics              | Value   | SYN Flood              |  | 0          |  |
| False Positives         | 0       | UDP Flood              |  | 0          |  |
| FP Rate                 | 0.00%   | Uncertain              |  | 3          |  |
| IF Precision            | 100.00% |                        |  |            |  |
| IF Recall               | 99.77%  | Action                 |  | Count      |  |
| IF F1-Score             | 99.89%  | Quarantined            |  | 3          |  |
| IF Accuracy             | 99.94%  | Time Ban               |  | 3          |  |
|                         |         | Blackhole              |  | 0          |  |
|                         |         | Released               |  | 0          |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 99.77%  | Share of actual attacks successfully flagged         |
| F1-Score                  | 99.89%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.94%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.23%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 99.77%  | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 4783           | FN 11             |
| Actual: Normal | FP 0              | TN 12689          |

### 2b. Random Forest: Attack Classification

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 100.00% | Share of actual attacks successfully flagged         |
| F1-Score                  | 100.00% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 100.00% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.00%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 100.00% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 0              | 0               | 0              |
| Act: ICMP | 0              | 4783            | 0              |
| Act: UDP  | 0              | 0               | 0              |

### 2c. Hybrid Cascaded

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 99.77%  | Share of actual attacks successfully flagged         |
| F1-Score                  | 99.89%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.94%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.23%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 99.77%  | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 4783           | FN 11             |
| Actual: Normal | FP 0              | TN 12689          |

2d. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 986.0 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 2.1 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2e. Controller Resource Overhead

| Metric                         | Value  | Description                                                           |
|--------------------------------|--------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 1.32%  | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 11.26% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 10.81% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.23 | 4        | 3            | 2026-09-26 19:28:58 | 2026-09-26 19:31:38 |
| 10.0.0.24 | 4        | 3            | 2026-09-26 19:28:57 | 2026-09-26 19:31:38 |
| 10.0.0.25 | 4        | 3            | 2026-09-26 19:29:01 | 2026-09-26 19:31:42 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action         |
|---------------------|-----------|-------|------------|------------|----------|----------------|
| 2026-09-26 19:28:57 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 19:28:58 | 10.0.0.23 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 19:29:01 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 19:29:08 | 10.0.0.24 | DDoS  | ICMP Flood | 90.7%      | High     | Time Ban (30s) |
| 2026-09-26 19:29:08 | 10.0.0.24 | DDoS  | ICMP Flood | 91.0%      | Critical | Time Ban       |
| 2026-09-26 19:29:09 | 10.0.0.23 | DDoS  | ICMP Flood | 90.8%      | High     | Time Ban (30s) |
| 2026-09-26 19:29:10 | 10.0.0.23 | DDoS  | ICMP Flood | 90.8%      | High     | Time Ban       |
| 2026-09-26 19:29:12 | 10.0.0.25 | DDoS  | ICMP Flood | 91.1%      | High     | Time Ban (30s) |
| 2026-09-26 19:29:14 | 10.0.0.25 | DDoS  | ICMP Flood | 91.1%      | Critical | Time Ban       |
| 2026-09-26 19:30:08 | 10.0.0.24 | DDoS  | ICMP Flood | 92.1%      | High     | Time Ban (1m)  |
| 2026-09-26 19:30:09 | 10.0.0.23 | DDoS  | ICMP Flood | 92.1%      | High     | Time Ban (1m)  |
| 2026-09-26 19:30:12 | 10.0.0.25 | DDoS  | ICMP Flood | 92.1%      | High     | Time Ban (1m)  |
| 2026-09-26 19:31:38 | 10.0.0.23 | DDoS  | ICMP Flood | 93.6%      | Critical | Time Ban (2m)  |
| 2026-09-26 19:31:38 | 10.0.0.24 | DDoS  | ICMP Flood | 91.5%      | Critical | Time Ban (2m)  |
| 2026-09-26 19:31:42 | 10.0.0.25 | DDoS  | ICMP Flood | 94.4%      | Critical | Time Ban (2m)  |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.25 | ICMP Flood | 0.6435   | 94.4% | Critical | Phase 2 | 1        | 10s      | 2026-09-26 19:31:42 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6437   | 91.5% | Critical | Phase 2 | 1        | 10s      | 2026-09-26 19:31:38 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6451   | 93.6% | Critical | Phase 2 | 1        | 10s      | 2026-09-26 19:31:38 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6418   | 92.1% | High     | Phase 2 | 3        | 1m 11s   | 2026-09-26 19:30:12 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6187   | 92.1% | High     | Phase 2 | 3        | 1m 11s   | 2026-09-26 19:30:09 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6278   | 92.1% | High     | Phase 2 | 3        | 1m 11s   | 2026-09-26 19:30:08 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6151   | 91.6% | High     | Phase 2 | 2        | 41s      | 2026-09-26 19:29:42 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6447   | 91.6% | High     | Phase 2 | 2        | 41s      | 2026-09-26 19:29:39 | Attack persisted - escalating to time ban |

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.24 | ICMP Flood | 0.6458   | 91.3% | High     | Phase 2 | 2        | 41s      | 2026-09-26 19:29:38 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6454   | 91.1% | High     | Phase 2 | 1        | 11s      | 2026-09-26 19:29:12 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6441   | 90.8% | High     | Phase 2 | 1        | 11s      | 2026-09-26 19:29:09 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6461   | 90.7% | High     | Phase 2 | 1        | 11s      | 2026-09-26 19:29:08 | Attack persisted - escalating to time ban |

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE